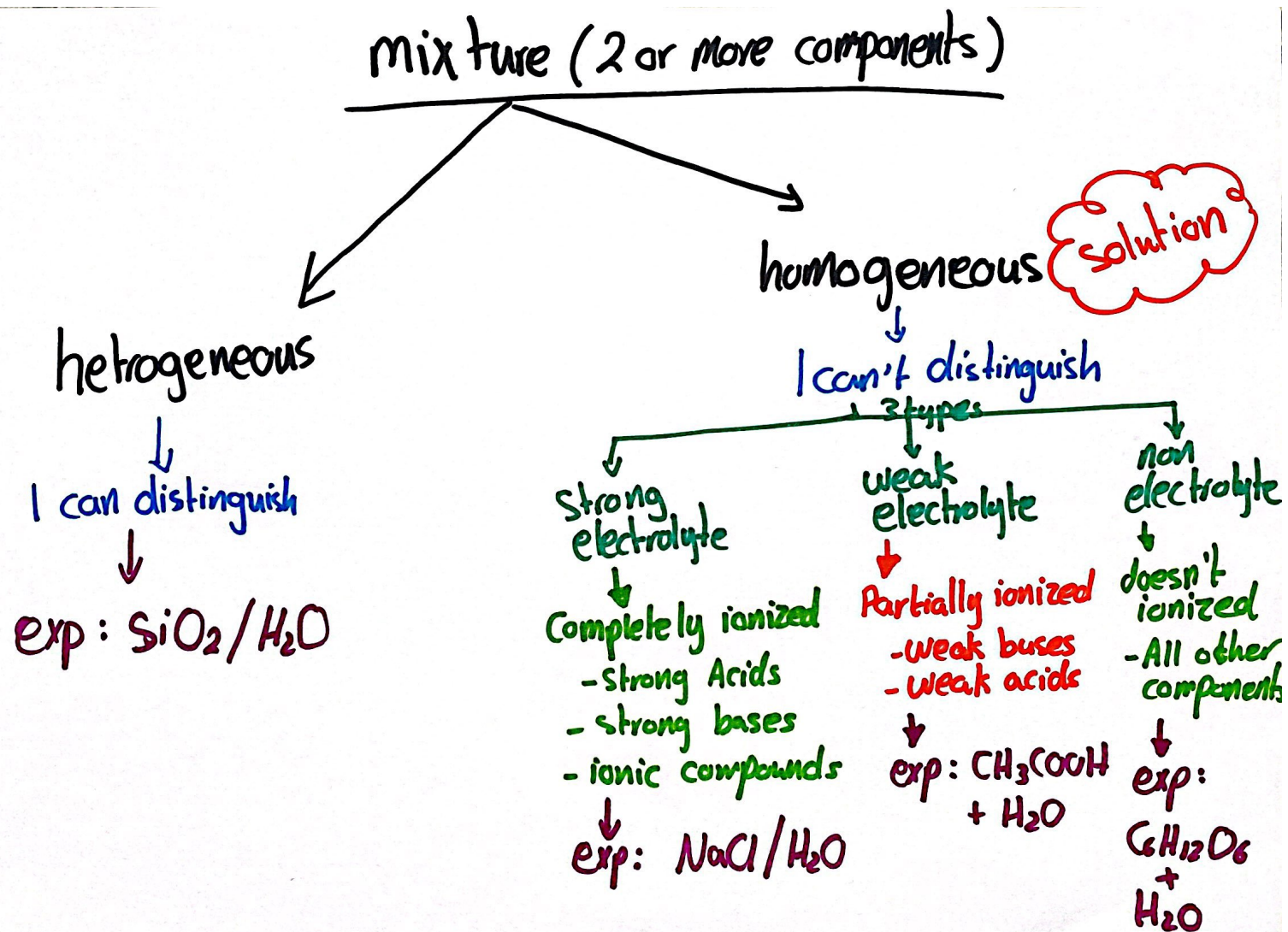


بسم الله الرحمن الرحيم
Ch.13 : Properties of Solutions

Solution = Solvent + Solute

↓
homogeneous mixture
of two or more
substances.

↓
the substance
present in the
greatest quantity.



1
Separation of
solute particles
 $\Delta H_{\text{solute}} > 0$

2
Separation of
solvent particles
to make holes
 $\Delta H_{\text{solvent}} > 0$

3
Formation of new
interactions between
solute and solvent.
 $\Delta H_{\text{mix}} < 0$

$$\Delta H_{\text{soln}} = \Delta H_{\text{solvent}}^+ + \Delta H_{\text{solute}}^+ + \Delta H_{\text{mix}}^-$$

$$(\Delta H_{\text{solv}} + \Delta H_{\text{solu}}) < \Delta H_{\text{mix}}$$

enthalpy favorable (exo)

exp: MgSO_4 (aq)

Solution:
 $\Delta S > 0$
 entropy favorable

$$(\Delta H_{\text{solv}} + \Delta H_{\text{solu}}) > \Delta H_{\text{mix}}$$

enthalpy unfavorable (endo)

exp: NH_4NO_3 (aq)

Why spon?

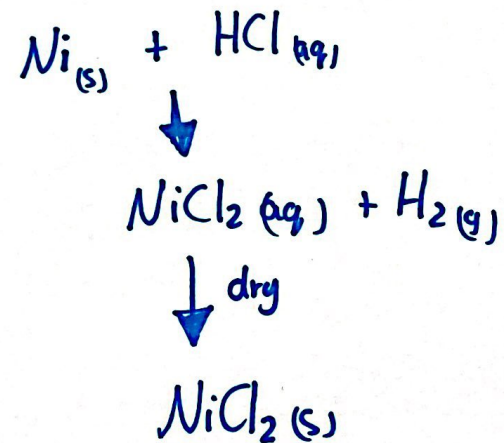
[Entropy Factor] Ch. 19

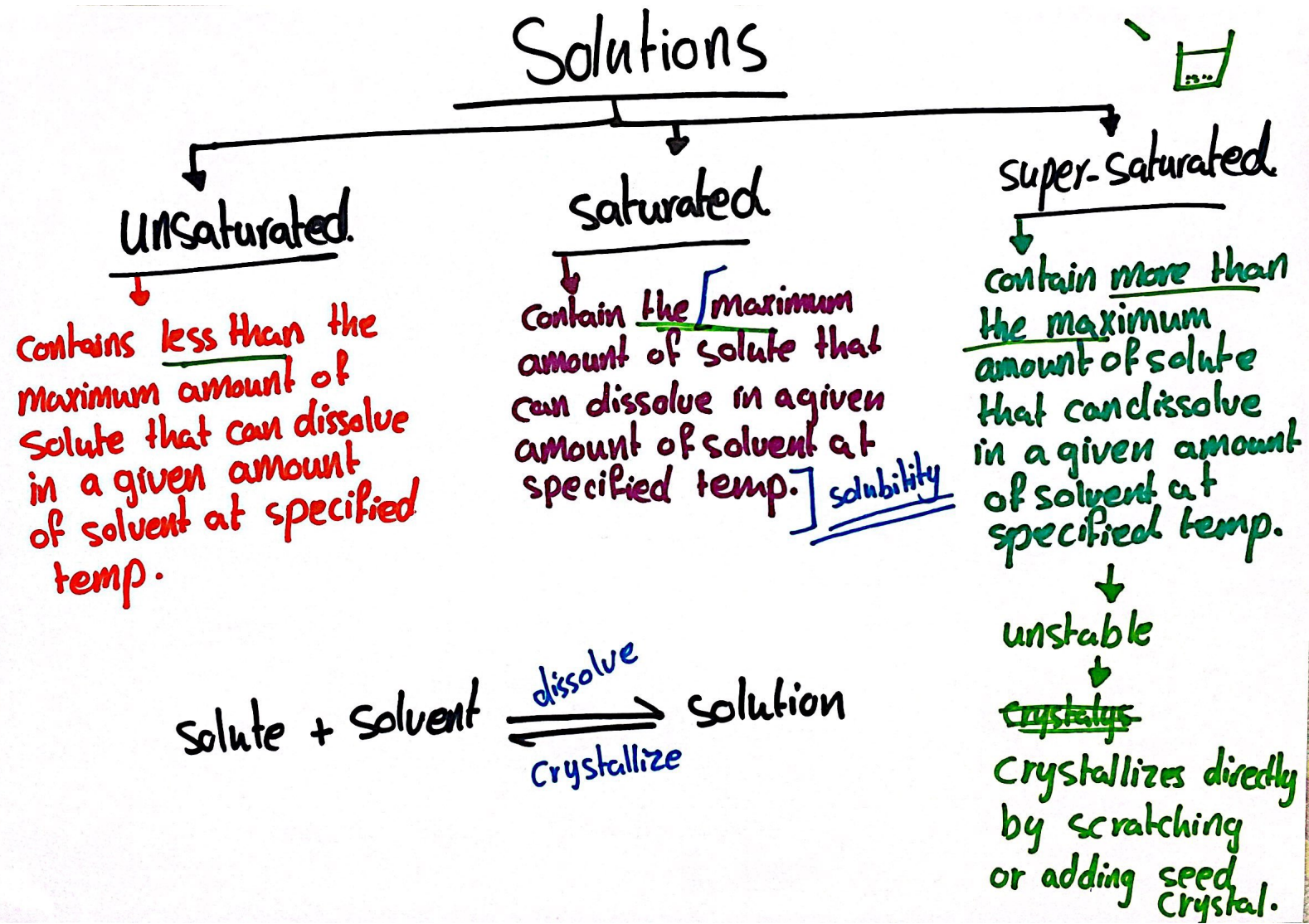
Solution Vs Chemical rxn

↙
Can get back
the original solute
by evaporating
the solvent.

⇓
Physical
change

↘
You can't get it
back.





Like Dissolve Like

Miscible

pairs of liquids
that mix in all
proportions.

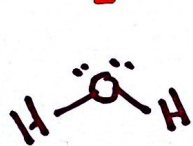
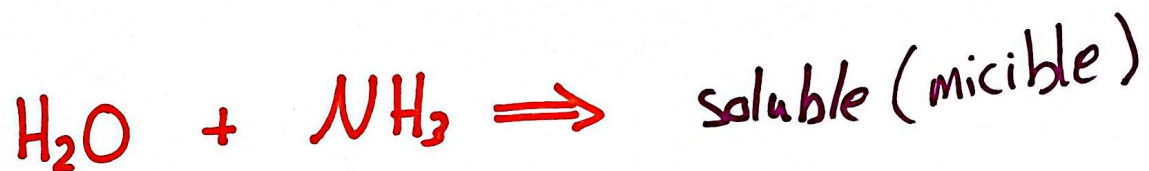
Soluble

Immiscible

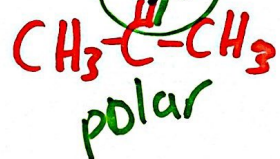
pairs of liquids that
don't dissolve in one
another.

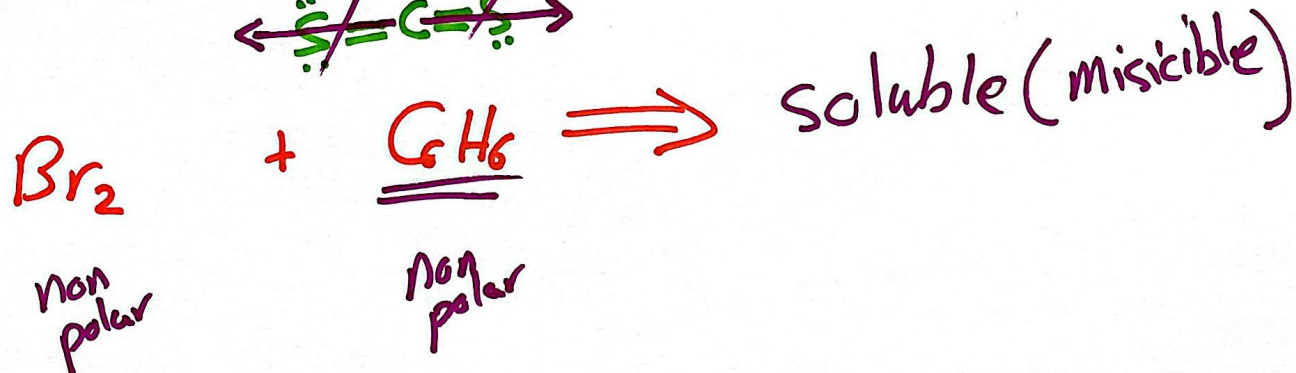
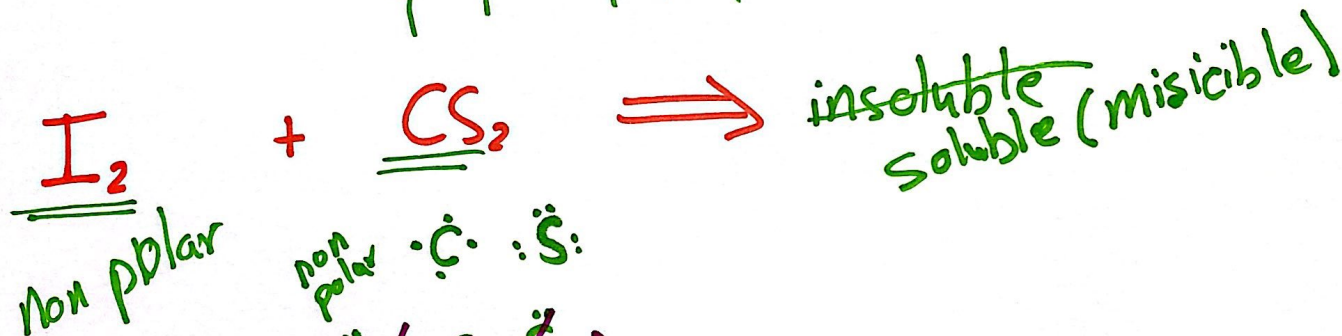
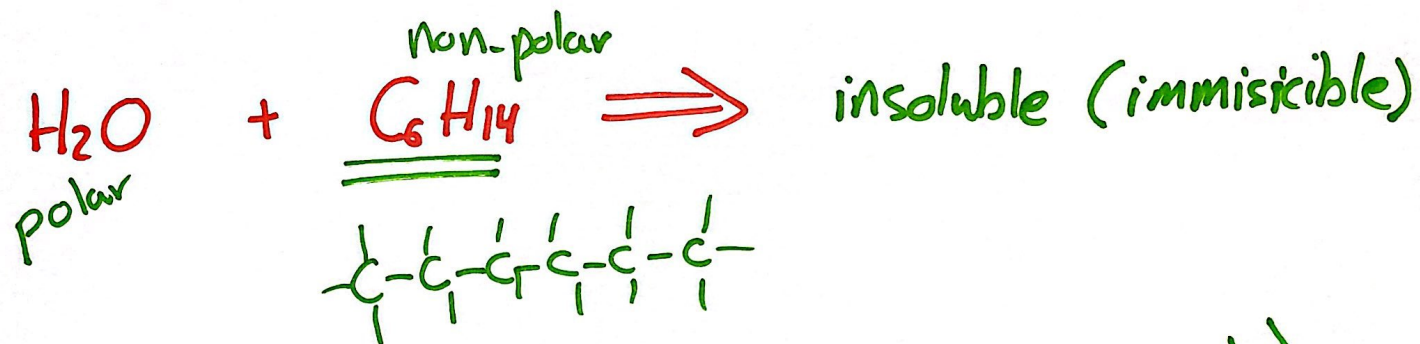
Insoluble

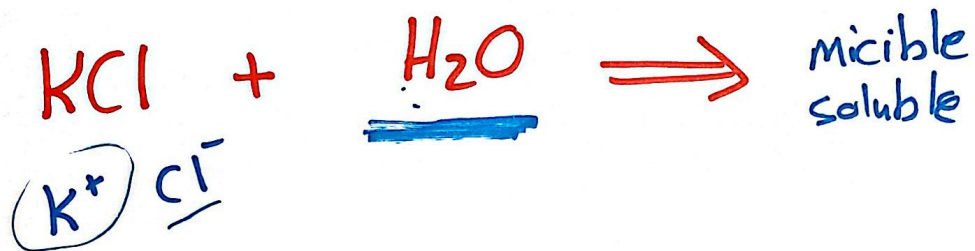
Examples



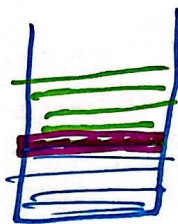
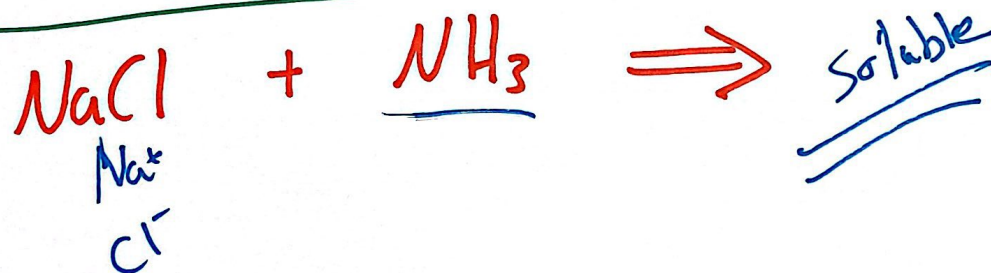
+







Hydration
the process in which an ion or molecule is surrounded by water molecules.



Solvation
the process in which an ion or a molecule is surrounded by solvent molecules



non polar

non polar

more soluble



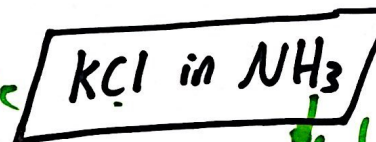
polar



polar



less soluble



polar



non polar

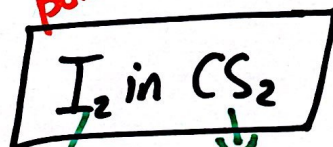


less soluble



polar

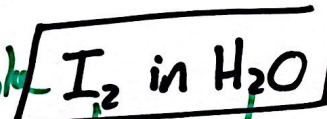
polar



non polar

non polar

more soluble



polar

General Rule $\approx$ Stronger attractions between solute and solvent $\Rightarrow$ greater solubility.

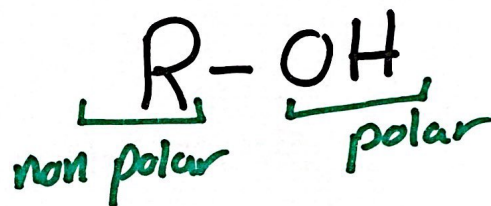
Solubility of gases in water inc with
inc mass.

Gas	M.m	solubility
O ₂	32	1.38×10^{-3}
Ar	39.9	1.5×10^{-3}
Kr	83.8	2.77×10^{-3}

↓ because
larger molecules have stronger
dispersion (London) forces.

N ₂	28	0.69×10^{-3}
CO	28	1.04×10^{-3}

Alcohol



R : hydrocarbon chain.

Exp : CH_3OH
 $\text{CH}_3\text{CH}_2\text{OH}$
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

Solubility in hexane inc. ↓

↑ solubility in water inc.

of C ↑ solubility in water ↓

Solubility in nonpolar ↑

Pressure effects

- Solids and Liquids $\Rightarrow$ independent
 - doesn't change -
- Gas : inc solubility with inc partial pressure.

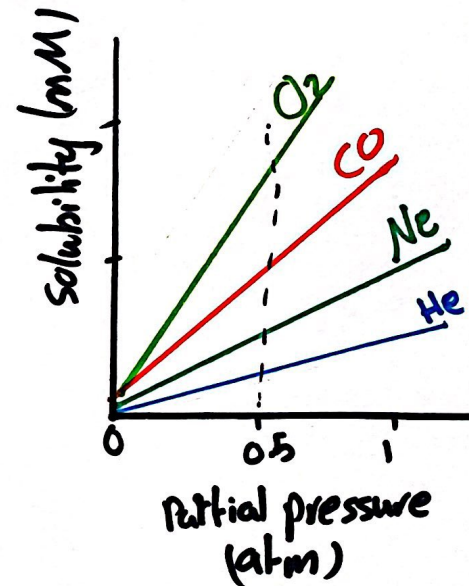
$$S_g = K P_g$$

Henry's Law

Solubility (M) $\swarrow$ Henry's Constant (M/atm) $\downarrow$ partial Pressure (atm) $\searrow$

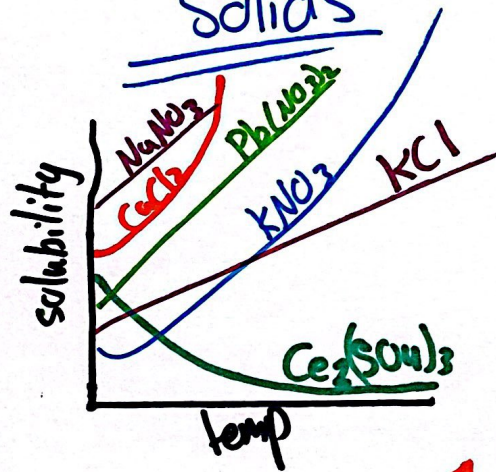
$\rightarrow$ depends on :

- solute
- solvent
- temp.



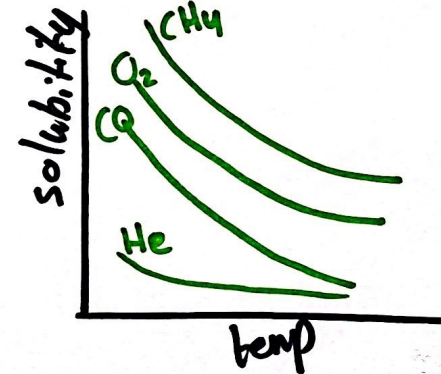
Temperature effects

Solids



Most solution formation
↓
Solubility ↑ temp ↑ ← **endo**
↓
↓ $[\text{Ce}_2(\text{SO}_4)_3 \text{ dec}]$ ← **exo**

Gases



Solubility ↓ temp ↑

- warm lake
- soft drinks